Properties

Physical properties

(i) Chloroform is a colourless liquid with peculiar sickly smell and a burning taste

(ii) The vapours of chloroform when inhaled it causes unconsciousness (depress the central nervous system) and hence it is used as an anaesthetic.

Chemical properties

1) Oxidation

Chloroform undergoes oxidation in the presence of light and air to form phosgene (carbonyl chloride)

$$\underset{\text{Chloroform}}{CHCl_3} + \frac{1}{2}O_2 \xrightarrow[\text{light}]{\text{air}} \underset{\text{Phosgene}}{COCl_2} + HCl$$

Since phosgene is very poisonous, its presence makes chloroform unfit for use as anaesthetic.

2) Reduction

Chloroform undergoes reduction with zinc and HCl in the presence of ethyl alcohol to form methylene chloride.

$$\underset{\text{Chloroform}}{CHCl_3} + 2[H] \xrightarrow[HCl]{Zn} \underset{\text{Methylene chloride}}{CH_2Cl_2} + HCl$$

3) Nitration

Chloroform reacts with nitric acid to form chloropicrin.(Trichloro nitro methane)

$$\underset{\text{Chloroform}}{CHCl_3} + HNO_3 \xrightarrow{\Delta} \underset{\text{Chloropicrin}}{CCl_3NO_2} + H_2O$$

It used as an insecticide and soil sterilising agent.

4) Carbylamine reaction

Chloroform reacts with aliphatic or aromatic primary amine and alcoholic caustic potash, to give foul smelling alkyl isocyanide (carbylamines)

$$\underset{\text{Methylamine}}{CH_3NH_2} + \underset{\text{Chloroform}}{CHCl_3} + 3KOH \longrightarrow \underset{\text{Methylisocyanide}}{CH_3NC} + 3KCl + 3H_2O$$

This reaction is used to test primary amine.

Evaluate Yourself

7) Chloroform is kept with a little ethyl alcohol in a dark coloured bottle why?

14.6.3 Tetra haloalkane

Carbon tetrachloride is a good example for tetra haloalkane

Carbon tetrachloride

Preparation

1. Chlorination of methane

The reaction of methane with excess of chlorine in the presence of sunlight will give carbon tetrachloride as the major product.

$$\underset{\text{Methane}}{CH_4} + 4Cl_2 \xrightarrow{h\nu} \underset{\text{Carbon tetrachloride}}{CCl_4} + 4HCl$$

Unit 14.indd 248 23-12-2021 17:22:57